Description

Illustration





Figure caption

1. Close-up of the material © John Fernandez 2. Granite Butts Mehre Heritage Hall, University of Georgia. © John Fernandez. Granite is most often used today in the form of thin slabs as a building cladding or in the interiors.

The material

Granite is one of the most durable of materials. The pyramid of Giza (2980 BC) was built of limestone cased in granite. It is a coarse grained igneous rock, very hard and compact. It takes a fine polish, showing the beauty of its crystals. It is the most important stone in building - whole cities are built of it (Aberdeen, in Scotland UK, is an example). It is used for work-surfaces because of its durability, resistance to acids and alkalis, and its decorative appearance. Granites, because of the lack of porosity, do not weather or crack as more porous stones do.

Compositional summary

Granite is made up of quartz (SiO2) and feldspars and micas (both complex alumino-silicates).

General properties

Density		2.63e3	-	3.2e3	kg/m^3
Price	*	0.785	-	4.7	GBP/kg
Date first used		-10000			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✓

Mechanical properties

Young's modulus		50	-	70	GPa
Shear modulus		21	-	29	GPa
Bulk modulus	*	30	-	45	GPa
Bending modulus	*	40	-	60	GPa
Poisson's ratio		0.15	-	0.26	
Yield strength (elastic limit)		8	-	23	MPa
Tensile strength		8	-	23	MPa
Compressive strength		110	-	255	MPa
Bending strength		10	-	20	MPa
Elongation	*	0.01	-	0.05	% strain
Hardness - Vickers		27	-	45	HV
Fatigue strength at 10^7 cycles	*	4	-	16	MPa
Fracture toughness		0.9	-	2.2	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.01	-	0.03	

Thermal and combustion properties

Thermal conductor or insulator?		Poor insulator			
Thermal resistivity		0.358	-	0.398	m.°C/W
Thermal expansion coefficient		10	-	12	µstrain/°C
Specific heat capacity		775	-	840	J/kg.°C
Melting point	*	1.2e3	-	1.6e3	°C
Maximum service temperature	*	600	-	700	°C
Minimum service temperature		-273			°C
Flammability		Non-flammable			
Emissivity		0.44			

Hygro-thermal properties

Water absorption		0.1	-	0.5	%
Frost resistance		Good			

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity	*	1e13	-	1e14	µohm.cm
Dielectric constant (relative permittivity)		14	-	18	
Dissipation factor (dielectric loss tangent)	*	0.001	-	0.01	
Dielectric strength (dielectric breakdown)	*	5	-	12	MV/m

Optical properties

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Natural color

A range of grays, but with occasional reds and greens.

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	No			

Acoustic properties

Sound absorption	Poor			
Sound isolation	Good			

Processability (scale 1 = impractical to 5 = excellent)

Machinability	2	-	3	

Durability

Water (fresh)	Excellent
Water (salt)	Excellent
Weak acids	Excellent
Strong acids	Acceptable
Weak alkalis	Excellent
Strong alkalis	Acceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Excellent
Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent

Primary material production: energy and climate change

Climate change (CO2-eq) (virgin grade production)	* 0.666	-	0.734	kg/kg
Embodied energy (virgin grade production)	* 10.5	-	11.6	MJ/kg
Water Usage	* 3.23	-	3.57	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 26.8	-	41.8	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.75	-	2.74	kg/kg

Recycling and end of life

Recycle	✗			
Functional recycle	✗			
Recycle fraction in current supply	1.34	-	1.48	%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Granite is exceptionally hard, making it expensive to quarry and work, but it is exceptionally durable and wear resistant, and can be polished to a mirror-like finish. Its great chemical stability gives resistance to aqueous and organic fluids, acids and alkalis. Granite is used in optical tables; streets paved with granite cobbles looked today as they did when they were laid. Stone weathers in a benign attractive way, unlike concrete; and its association with the great structures of the past make it attractive for imposing buildings of today.

Technical notes

Granites are made up of crystals of feldspar, dull white to deep red in color, with small grains of gray quartz and crystals of mica that give the stone its sparkle.

Typical uses

Granites are used for building and facing, floors and work surfaces, stable supports for machine tools, optical tables and supports for other precision instruments. Much is now used in ornamental ways.

Related standards

04400 Stone
04410 Rough Stone
04420 Cut Stone
04440 Flagstone
04450 Stone Veneer
04465 Granite

Links

ProcessUniverse

Reference